Cover letter

12 October 2026

Editors, *Journal of Manufacturing Processes*

Dear Editors,

Please consider our manuscript, "Reinforcement Learning-Based Computer Numerical Control Motion Planning with Adaptive Preview and Jerk-Constrained Action Projection", for publication as a research article.

Consecutive corners in CNC toolpaths restrict both transition geometry and achievable feedrate. Our manuscript presents an offline learned planner that jointly selects steering, feedrate and preview. Planar kinematic projection constrains the resulting commands. The planner then refines startup and braking while preserving intermediate commands and checking the complete trajectory.

The simulation study compares four selected complete-path cases, comprising two development programs and two enlarged synthetic cases, under common geometric tolerances and planar kinematic limits. The command trajectories explain how executed path length and accumulated speed deficit contribute to traversal time.

The work addresses path shaping and constrained motion generation in CNC process planning, within the manufacturing-process scope of the Journal of Manufacturing Processes.

Sincerely,

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